YEAR 7 · UNIT 4 · EXTRA HOMEWORK

Where the Point Goes

Comparing · adding and subtracting · multiplying and dividing · the words that choose

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 23 October 2026

Why this sheet exists

There is **nothing new to calculate** in Unit 4. 3.2×7 is 32×7, and you already know how to do that. $8.24 \div 4$ is short division. The one new question is **where the point goes**.

Every question on this sheet prints its steps as **boxes**. Fill the boxes — that *is* the method. An **orange box** means: write a **0** here.

No calculator.

Part 1 · Which is bigger?

X1 Two parts of a number

Write the **whole-number part** and the **decimal part**. The first one is done for you.

Number	Whole-number part	Decimal part
3.25	3	.25
14.6		
0.38		
7.05		
120.9		

X2 Fill with zeros, then compare

Compare decimals

First look at the **whole-number parts**. If they are the same, **fill with zeros** so both numbers have the same number of digits after the point. Then compare. 3.3 = 3.30, and 3.30 > 3.25.

Write a 0 in each orange box. Then write the **bigger** number. The first one is done for you.

The two numbers	Fill with zeros	Bigger
3.3 and 3.25	3.30 and 3.25	3.3
0.6 and 0.52	0.6 ☐ and 0.52	
4.1 and 4.09	4.1 ☐ and 4.09	
2.75 and 2.8	2.75 and 2.8 ☐	
6.07 and 6.7	6.07 and 6.7 ☐	
0.5 and 0.499	0.5 ☐☐ and 0.499	

X3 Put them in order

Step 1. Fill with zeros, so every number has three digits after the point.

2.4☐☐ 2.38☐ 2.405 2.04☐

Step 2. Write the four numbers **in order of size, starting with the smallest**.

_____ _____ _____ _____

X4 The snail race

Three snails race 1 metre across Mr Bowen's desk. These are their times.

Word help

fastest = the **smallest** time. **slowest** = the **biggest** time.

Snail	Time (min)	Fill with zeros
A	8.5	8.5☐
B	8.25	8.25
C	8.05	8.05

(a) Which snail was the **fastest**? _____

(b) Which snail was the **slowest**? _____

Part 2 · Line up the points

Adding and subtracting

Write the numbers with the **points in a line**. Fill the gaps with **zeros**. Work from the right, like whole numbers. The point in the answer goes **straight down**.

X5 The point is already there

Write a 0 in every orange box. Then add or subtract. The first one is done for you.

(a) 4.6 + 2.35

		.		
	4	.	6	0
+	2	.	3	5
	6	.	9	5

(b) 12.4 + 3.08

		.		
1	2	.	4	
+	3	.	0	8
		.		

(c) 7 + 0.45

		.		
	7	.		
+	0	.	4	5
		.		

(d) 8.5 − 2.36

		.		
	8	.	5	
−	2	.	3	6
		.		

(e) 10 − 3.4

		.		
1	0	.		
−	3	.	4	
		.		

(f) 1 − 0.36

		.		
	1	.		
−	0	.	3	6
		.		

(g) In **(c)**, where is the point in the number 7?

_____

X6 Change

Word help

change: in a shop, the money you get **back** when you pay too much. 10 dollars is written **10.00**.

(a) Mr Bowen pays for a 6.45-dollar book with a 10-dollar note. How much change does he get?

		.		
1	0	.		
−	6	.	4	5
		.		

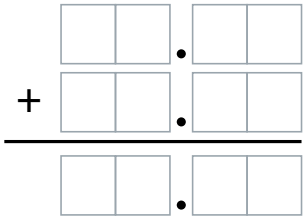
(b) Mr Bowen pays for 12.80-dollar shoes with a 20-dollar note. How much change does he get?

		.		
2	0	.		
−	1	2	.	8	0
		.		

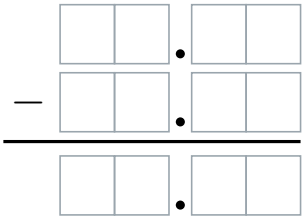
X7 Now you line them up

Write the numbers in the grid yourself. The point goes in the point column. Write a 0 in any gap after the point. Then work it out.

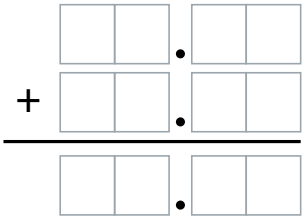
(a) 5.07 + 12.6



(b) 9 − 4.25



(c) 0.8 + 3.45



Part 3 · Point out, point back

Multiplying a decimal

3.2 × 7. Take the point out: 32 × 7 = 224. There is **1 decimal place** in 3.2, so put **1** back: 22.4.
Estimate to check: 3 × 7 = 21. 22.4 is close to 21. ✓

X8 Four steps

Fill in the table. The first one is done for you.

Calculation	Point out	Decimal places	Point back	Estimate
2.4 × 3	24 × 3 = 72	1	7.2	2 × 3 = 6
1.6 × 4	16 × 4 =			2 × 4 =
0.9 × 6	9 × 6 =			1 × 6 =
3.25 × 2				
5.3 × 7				
4.05 × 3				

X9 The estimate decides

All three answers have the same digits. Work out the estimate, then **circle** the answer that is close to it.

Calculation	Estimate	Circle one		
6.8 × 4	7 × 4 =	2.72	27.2	272
3.9 × 6		2.34	23.4	234
12.5 × 3		3.75	37.5	375
0.8 × 9		0.72	7.2	72

X10 Lighter on the Moon

On the Moon, weight = mass × **1.6**. On Mars, weight = mass × **3.7**. Fill in the steps.

(a) A 5 kg bag of rice on the **Moon**: 5×1.6

Point out: $5 \times 16 =$ _____ Point back: _____ N

(b) A 3 kg cat on the **Moon**: 3×1.6

Point out: $3 \times 16 =$ _____ Point back: _____ N

(c) A 7 kg bag on **Mars**: 7×3.7

Point out: _____ Point back: _____ N

Part 4 · The point goes straight up

Dividing a decimal

Use short division. The point in the answer goes **straight up**.
If a digit is too small to divide, write **0** in the answer and carry.
Check with the **inverse calculation**: $8.24 \div 4 = 2.06$, because $2.06 \times 4 = 8.24$.

X11 One digit in each box

The point is already in the answer. Write one digit in each box. In **(f)**, write a 0 in the orange box first. Then check.

(a) $6.84 \div 2$

$$2 \overline{)\, \square\square . \square\square \;\; 6.84}$$

Check: _____ × 2 = 6.84

(b) $9.45 \div 3$

$$3 \overline{)\, \square\square . \square\square \;\; 9.45}$$

Check: _____ × 3 = 9.45

(c) $4.28 \div 4$

$$4 \overline{)\, \square\square . \square\square \;\; 4.28}$$

Check: _____ × 4 = 4.28

(d) $7.2 \div 8$

$$8 \overline{)\, \square . \square \;\; 7.2}$$

Check: _____ × 8 = 7.2

(e) $5.6 \div 7$

$$7 \overline{)\, \square . \square \;\; 5.6}$$

Check: _____ × 7 = 5.6

(f) $2.6 \div 4$

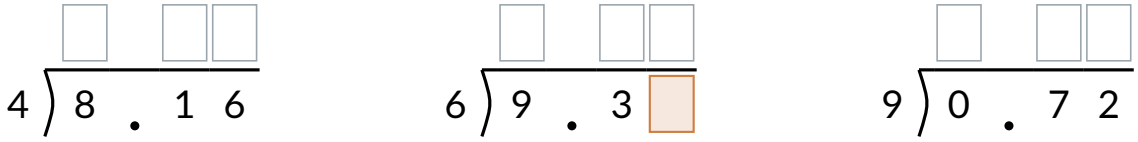
$$4 \overline{)\, \square\square . \square\square \;\; 2.6\square}$$

Check: _____ × 4 = 2.6

X12 Now you put the point up

This time the point is **not** in the answer. Put it there yourself — straight up. Then fill the boxes.

(a) 8.16 ÷ 4 **(b)** 9.3 ÷ 6 **(c)** 0.72 ÷ 9



X13 The mean, in two steps

The mean

Step 1: add up all the values. **Step 2:** divide by how many there are.

(a) Mr Bowen drops a ball three times and times it. The times are **0.46 s**, **0.44 s** and **0.45 s**. Find the mean.

Step 1: 0.46 + 0.44 + 0.45 = _____

Step 2: _____ ÷ 3 = _____ s

(b) His reaction times in the ruler catch are **0.21 s**, **0.19 s**, **0.23 s** and **0.17 s**. Find the mean.

Step 1: _____ = _____

Step 2: _____ ÷ ☐ = _____ s

Part 5 · Making it easier

X14 "Of" means ×

Half, a quarter, a tenth

half of = ÷ 2 = × 0.5 **a quarter of** = ÷ 4 = × 0.25 **a tenth of** = ÷ 10 = × 0.1

Fill in the table. The first one is done for you.

In English	Calculation	Answer
half of 18	18 ÷ 2	9
a quarter of 20	20 ÷ ☐	
a tenth of 35	35 ÷ ☐	
half of 7		
a quarter of 6		

X15 Times 9 and times 11

Start from × 10

× 9: do × 10, then **take away** one lot. × 11: do × 10, then **add** one lot.

(a) 2.6 × 9 = 26 − 2.6 = _____

(b) 1.4 × 9 = _____ − _____ = _____

(c) 2.6 × 11 = 26 + 2.6 = _____

(d) 3.2 × 11 = _____ + _____ = _____

X16 Split the number

Partitioning means splitting a number into easy parts: 4.3 = 4 + 0.3.

(a) 4.2 × 3 = 4 × 3 + 0.2 × 3 = _____ + _____ = _____

(b) 5.3 × 4 = 5 × 4 + 0.3 × 4 = _____ + _____ = _____

(c) 7.4 × 2 = _____ + _____ = _____

Part 6 · The words that choose

X17 Which operation?

Write +, −, × or ÷ for each one. The first one is done for you.

The words	Operation
altogether	+
change	
share equally	
how much do 4 cost?	
how much heavier?	
the total	

X18 Calculation first

Write the calculation in the boxes: number, sign, number. Then the answer.

(a) A bottle of water costs 0.85 dollars. Mr Bowen buys 4. How much does he pay?

_____ ☐ _____ = _____ dollars

(b) Mr Bowen pays for a 2.65-dollar juice with a 5-dollar note. How much change does he get?

_____ ☐ _____ = _____ dollars

(c) Mr Bowen and three friends share 9.20 dollars equally. How much does each person get?

How many people? ☐ _____ ☐ _____ = _____ dollars

(d) A bag of sugar has a mass of 0.5 kg. What is its **weight** on Earth? Weight = mass × 10.

_____ ☐ _____ = _____ N

X19 The whole sheet in one question

Mr Bowen buys 3 mangoes at 1.45 dollars each. He pays with a 5-dollar note. How much change does he get?

Step 1. The mangoes: 145 × 3 = _____, so 1.45 × 3 = _____ dollars.

Step 2. The change: 5.00 − _____ = _____ dollars.

Work it out here. Write 5 as 5.00 and line up the points.

Answer Key — Teacher Copy

X1 Two parts of a number

14.6: 14 and .6 · 0.38: 0 and .38 · 7.05: 7 and .05 · 120.9: 120 and .9

0.38 is the row that gets a blank: there is a whole-number part, and it is 0. 7.05 is the row that gets ".5". Both are the same habit — not writing a zero that is holding a place — and X2 depends on it.

X2 Fill with zeros, then compare

0.60, bigger 0.6 · 4.10, bigger 4.1 · 2.80, bigger 2.8 · 6.70, bigger 6.7 · 0.500, bigger 0.5

In every row the bigger number is the shorter one, and that is on purpose: it is the opposite of what "more digits" suggests, and the zeros are what show it.

6.07 against 6.7 is the row to discuss: both have a 0 and a 7, and it is *where* the 0 is that decides. The 0 in 6.07 holds the tenths open; the 0 in 6.70 adds nothing.

0.5 against 0.499 needs two boxes. A student who writes 0.50 and compares 50 with 499 has still been fooled by length.

X3 Put them in order

Step 1: 2.400, 2.380, 2.405, 2.040
Step 2: 2.04, 2.38, 2.4, 2.405

Accept the numbers written with or without their filled-in zeros.

2.4 against 2.405 is the close pair: 2.400 < 2.405, so 2.4 comes first. Expect 2.405 before 2.4 from a student who stopped filling at two places. 2.04 first is the one most will get, because the zero is so visible.

X4 The snail race

8.50 · (a) **C** (8.05) · (b) **A** (8.5)

The slowest snail has the shortest-looking time. Expect B for (b) from anyone comparing length, and A for (a) from anyone who has read "fastest" as "biggest". The word help is on the page; send them back to it.

X5 The point is already there

(b) 12.40 + 3.08 = **15.48** (c) 7.00 + 0.45 = **7.45** (d) 8.50 − 2.36 = **6.14**
(e) 10.0 − 3.4 = **6.6** (f) 1.00 − 0.36 = **0.64**
(g) After the 7: 7 = 7.00. (Accept "at the end", "after the 7".)

The grid makes the alignment impossible to get wrong, so what is left is the subtraction with zeros. (d) and (f) need a borrow from a 0 the student wrote themselves; expect 6.26 for (d) and 1.36 for (f) — "take the smaller digit from the bigger" in each column, which a 0 on top invites.

(g) is the question the grid answers by itself. A whole number has its point at the end, and that is why (c)'s 7 is in the ones column and not under the 4.

X6 Change

(a) 10.00 − 6.45 = **3.55** dollars (b) 20.00 − 12.80 = **7.20** dollars

(b) answers in money, so 7.20, not 7.2 (HW 9's E3).

(a) borrows across three zeros and is the hardest subtraction on the sheet. If it goes wrong, check it column by column from the right; the inverse (3.55 + 6.45 = 10.00) is the fastest way for the student to see it.

X7 Now you line them up

(a) 5.07 + 12.60 = **17.67** (b) 9.00 − 4.25 = **4.75** (c) 0.80 + 3.45 = **4.25**

Look at where the digits went before marking the answer. (a) is the test: 12.6 needs the tens column and 5.07 does not, so a student who fills the grid from the left puts the 5 under the 1 and gets 62.67 or similar.
(b): the 9 goes in the ones, with two zeros after the point. Expect 9 − 4.25 written as 9.25 − 4 or with the 9 in the hundredths.

X8 Four steps

1.6 × 4	16 × 4 = 64	1	6.4	2 × 4 = 8
0.9 × 6	9 × 6 = 54	1	5.4	1 × 6 = 6
3.25 × 2	325 × 2 = 650	2	6.50 = 6.5	3 × 2 = 6
5.3 × 7	53 × 7 = 371	1	37.1	5 × 7 = 35
4.05 × 3	405 × 3 = 1215	2	12.15	4 × 3 = 12

The third column is the step, and it counts decimal places in the *question*, not digits. Expect 3 for 3.25 from anyone counting all the digits; the answer then comes out 0.650 and the estimate of 6 throws it out.
4.05: the 0 is a digit and stays in when the point comes out — 405, not 45. Expect 45 × 3 = 135, 1.35, from anyone who drops it. The estimate (12) catches that too.

X9 The estimate decides

27.2 (7 × 4 = 28) · 23.4 (4 × 6 = 24) · 37.5 (12 × 3 = 36, or 13 × 3 = 39) · 7.2 (1 × 9 = 9)

No multiplication is needed in this question and that is the point of it. A student who works out 68 × 4 first has done more than was asked and should be told so — warmly.
0.8 × 9: the estimate 1 × 9 = 9 is a bit big, and 7.2 is the only choice near it. Some will circle 0.72 because "0.8 is less than 1"; ask whether nine lots of nearly 1 could be less than 1.

X10 Lighter on the Moon

(a) 80, so **8** N (8.0) (b) 48, so **4.8** N (c) 7 × 37 = 259, so **25.9** N

(a) comes back as 8.0, which is 8; accept both (nothing was rounded). Expect 80 from a student who skipped putting the point back — a bag of rice weighing 80 N on the Moon would weigh more than on Earth (50 N), which is worth asking about.
This is HW 9's F2 with smaller numbers and the steps printed.

X11 One digit in each box

(a) 3.42 · (b) 3.15 · (c) **1.07** · (d) **0.9** · (e) **0.8** · (f) **0.65**
Checks: 3.42 × 2, 3.15 × 3, 1.07 × 4, 0.9 × 8, 0.8 × 7, 0.65 × 4.

(c), (d) and (e) each need a 0 written in an answer box. The frame has a box there, so a blank box is visible — that is the scaffold doing its job. Expect 1.7 for (c) if the student skipped the box.
(d) and (e) start with "8 into 7 will not go": the 0 goes in the ones, then the remainder carries.
(f) runs out of digits: 2.60, and the orange 0 is where the last 5 comes from. Expect "0.6 remainder 2".

X12 Now you put the point up

(a) **2.04** (b) 9.30 ÷ 6 = **1.55** (c) **0.08**

Look for the point in the answer row first; it should sit exactly above the dividend's point.
(c) is the hardest item in Part 4: 0 ÷ 9 = 0, then 7 ÷ 9 will not go, so a second 0, then 72 ÷ 9 = 8. Expect 0.8 and 8.
(b) needs the orange 0 to finish: 9.3 ÷ 6 stops at 1.5 remainder 3, and 30 ÷ 6 = 5 gives the last digit.

X13 The mean, in two steps

(a) 0.46 + 0.44 + 0.45 = 1.35, 1.35 ÷ 3 = **0.45** s
(b) 0.21 + 0.19 + 0.23 + 0.17 = 0.80, 0.80 ÷ 4 = **0.2** s

(a) is the drop time from HW 9's F3, so a student doing both sheets meets 0.45 s twice — on purpose. (b): "how many there are" is 4, the box in Step 2. Expect ÷ 2 from a student who counted the digits after the point. The answer, 0.2 s, is the reaction time the Science 3.1 deck quoted.

X14 "Of" means ×

a quarter of 20: 20 ÷ 4 = 5 · a tenth of 35: 35 ÷ 10 = 3.5 · half of 7: 7 ÷ 2 = 3.5 · a quarter of 6: 6 ÷ 4 = 1.5

The last two rows have no calculation printed; a student who writes only the answer has skipped the column.
"A quarter of 6" is the hardest: 6 ÷ 4 does not divide exactly, so it is short division with the point — 6.0 ÷ 4 = 1.5 — which is Part 4 again.

X15 Times 9 and times 11

(a) 23.4 (b) 14 − 1.4 = 12.6 (c) 28.6 (d) 32 + 3.2 = 35.2

(a) and (b) are subtractions with a hidden zero: 26.0 − 2.6. Expect 24.6 from 26 − 2 + 0.6: the 0.6 added instead of taken away. Writing 26.0 in a column fixes it.

X16 Split the number

(a) 12 + 0.6 = 12.6 (b) 20 + 1.2 = 21.2 (c) 7 × 2 + 0.4 × 2 = 14 + 0.8 = 14.8

0.3 × 4 = 1.2 is the one to watch: it crosses 1. Expect 0.12, which is a decimal place counted in the wrong place.

X17 Which operation?

change − · share equally ÷ · how much do 4 cost? × · how much heavier? − · the total +

"How much do 4 cost?" is ×, and some will write + — which is not wrong, just slow (repeated addition). Accept it with a word.

X18 Calculation first

(a) 0.85 × 4 = 3.40 dollars · (b) 5.00 − 2.65 = 2.35 dollars
(c) **4** people; 9.20 ÷ 4 = 2.30 dollars · (d) 0.5 × 10 = 5 N

(c) is HW 9's E2(c) with the box that makes the student count: "Mr Bowen and three friends" is four. A 3 in the people box is the English error, and this sheet catches it before the division starts.
(d) is the one science question on the sheet, and the number comes from Science 3.1: $W = m \times 10$.

X19 The whole sheet in one question

Step 1: 145 × 3 = 435, so **4.35** dollars. Step 2: 5.00 − 4.35 = **0.65** dollars.

Point out and back (Part 3), then zeros written and subtracted (Part 2), with "change" choosing the operation (Part 6). Expect 1.35 from a student who subtracted 4.35 from 5.00 the easy way round in the tenths (X5(d)'s error).